

From Individual Grade Anomaly to Institutional Model Risk

A Quantitative Framework for the
Audit of Historical Assessment,
Fixed-Income Modelling, and
Scientific Governance

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Abstract

This paper develops a multidisciplinary framework for investigating a historical academic assessment anomaly involving a quantitative fixed-income securities course at BI Norwegian Business School. The central empirical observation is an F grade appearing within an otherwise substantially stronger transcript whose reported overall average is B. The paper does not infer from this observation that the grade was improperly awarded. Instead, it formalizes the observation as a statistical outlier and develops a sequence of increasingly demanding tests: transcript-level anomaly detection, course-level comparison, cohort-level analysis, assessment-process reconstruction, model-risk analysis, and institutional quality assurance.

Particular attention is given to the historical GRA 6539 Fixed Income Securities curriculum, which explicitly included interest-rate modelling, fixed-income options, binomial trees, Monte Carlo simulation, Excel-based quantitative work, and analysis of problems extending beyond explicitly covered material. The paper therefore investigates whether the computational methods available within the stated pedagogical framework could have been applied to propositions concerning path-dependent fixed-income securities and interest-rate models.

The paper also distinguishes the mathematical limitations of the Ho–Lee model from claims that the model is incapable of pricing every path-dependent claim. A rigorous formulation shows that a Ho–Lee lattice can be augmented with a barrier-state variable, while also showing that Ho–Lee alone does not specify every additional stochastic factor required by arbitrary path-dependent claims. The resulting framework connects statistics, finance, numerical analysis, computer science, risk management, scientific governance, institutional economics, and quality assurance.

No causal conclusion concerning the disputed grade is asserted without the contemporaneous assessment record. The paper instead proposes an auditable evidence hierarchy in which the primary unresolved question is: what contemporaneous evidence exists showing how the historical assessment was actually performed and how the final grade was determined?

The paper ends with “The End”

1 Introduction

Academic grades are normally treated as observations generated by an assessment system. In quantitative disciplines, however, an individual grade can also be regarded as an observation in a statistical process. An unusually low observation does not prove that the underlying process is defective, but it can justify additional investigation.

The case considered here has an unusually useful structure. A student’s transcript contains several A and B grades, a number of C grades, two D grades, and one F. The transcript reports an overall average of B. The F is associated with GRA 6539 Fixed Income Securities, a course whose historical documentation explicitly included interest-rate modelling, fixed-income options, binomial trees, Monte Carlo simulation, and quantitative analysis.

The research question is therefore not merely whether an F is surprising. The question is whether the F is statistically anomalous, whether the course itself exhibits an unusual grade distribution, whether

the assessment process can be reconstructed, and whether the mathematical proposition being assessed was treated consistently with the course's stated quantitative objectives.

The investigation is structured as follows:

$$\text{Transcript} \longrightarrow \text{Statistical anomaly} \longrightarrow \text{Course comparison} \longrightarrow \text{Assessment reconstruction} \longrightarrow \text{Institutional audit}. \quad (1)$$

This ordering is important because each stage requires stronger evidence than the previous one.

2 The Evidentiary Principle

The fundamental logical distinction is between an observation, an anomaly, a causal explanation, and an institutional finding.

$$\boxed{\text{Observed anomaly} \not\Rightarrow \text{improper assessment}} \quad (2)$$

Similarly,

$$\boxed{\text{improper assessment} \not\Rightarrow \text{institution-wide misconduct}} \quad (3)$$

A defensible investigation must therefore distinguish four propositions:

$$H_1 : \text{The student's F is an unusually low observation relative to the student's record.} \quad (4)$$

$$H_2 : \text{The F is unusually low relative to the GRA 6539 cohort distribution.} \quad (5)$$

$$H_3 : \text{The assessment process contained a material procedural or academic error.} \quad (6)$$

$$H_4 : \text{The observed problem reflects a systematic institutional pattern.} \quad (7)$$

Evidence sufficient for H_1 is not automatically sufficient for H_2 , and evidence sufficient for H_2 is not automatically sufficient for H_3 or H_4 .

3 Historical GRA 6539 Context

The historical BI course documentation is particularly relevant because the course was not presented as a purely qualitative fixed-income course. The 2013–2014 and 2014–2015 documentation describes quantitative techniques including binomial-tree and Monte Carlo approaches, interest-rate modelling, fixed-income options, and Excel-based computational work.

The course documentation also states that students should be capable of analysing problems extending beyond those explicitly covered in class or the required book.

This yields the following methodological relationship:

$$\boxed{\text{GRA 6539} \supset \{\text{interest-rate modelling, fixed-income options, binomial trees, Monte Carlo}\}.} \quad (8)$$

The historical course documentation therefore establishes that quantitative simulation and lattice methods belonged to the methodological environment of the course.

It does *not*, however, establish that such a calculation was actually performed on any particular student's submission.

That distinction is central.

4 Transcript-Level Statistical Anomaly

Consider the numerical coding

$$A = 5, \quad B = 4, \quad C = 3, \quad D = 2, \quad E = 1, \quad F = 0. \quad (9)$$

For the transcript considered here, the grade counts are:

Table 1: Observed transcript grade composition

Grade	Number of courses
A	3
B	3
C	4
D	2
F	1

The resulting numerical mean is

$$\bar{G} = \frac{3(5) + 3(4) + 4(3) + 2(2) + 1(0)}{13} = \frac{43}{13} \approx 3.308. \quad (10)$$

The population standard deviation is approximately

$$\sigma_G \approx 1.380. \quad (11)$$

The standardized location of the F is consequently

$$Z_F = \frac{0 - \bar{G}}{\sigma_G} \approx -2.40. \quad (12)$$

Using the sample rather than population standard deviation produces a value of approximately

$$Z_F \approx -2.30. \quad (13)$$

Thus the F is approximately 2.3–2.4 standard deviations below the mean of the student’s own numerical transcript distribution.

This is a legitimate statistical anomaly.

It is not, by itself, evidence of grading misconduct.

5 Why the Z_F Statistic Matters

The statistic is useful because it transforms an intuitive observation into a formal testable proposition.

The relevant distinction is:

$$\boxed{Z_F \approx -2.4 \implies \text{unusually low transcript observation}} \quad (14)$$

rather than

$$\boxed{Z_F \approx -2.4 \implies \text{improper F.}} \quad (15)$$

The first implication is statistical.

The second would require additional evidence.

The appropriate next statistic is therefore a course-conditional statistic:

$$Z_{\text{course}} = \frac{G_i - \mu_{\text{GRA6539},t}}{\sigma_{\text{GRA6539},t}}, \quad (16)$$

where t denotes the relevant historical semester.

For the disputed assessment, t should be the actual semester of the student’s assessment rather than a contemporary semester.

6 Course-Level Anomaly Analysis

Suppose the historical GRA 6539 cohort contains n students and grade observations $G_1, \dots, G_n$. Define

$$\mu_{6539} = \frac{1}{n} \sum_{i=1}^n G_i, \quad (17)$$

$$\sigma_{6539}^2 = \frac{1}{n-1} \sum_{i=1}^n (G_i - \mu_{6539})^2. \quad (18)$$

The student's standardized course-level result is then

$$Z_{6539} = \frac{G_{\text{student}} - \mu_{6539}}{\sigma_{6539}}. \quad (19)$$

This statistic is more informative than Z_F because it conditions on the actual course population. A further benchmark is a matched-course comparison:

$$\Delta_{\text{course}} = \mu_{6539} - \mu_{\text{benchmark}}, \quad (20)$$

where the benchmark can consist of comparable fixed-income, derivatives, or financial-modelling courses from the same institutional environment.

The relevant comparison is therefore:

$$\boxed{\text{student transcript} \leftrightarrow \text{GRA 6539 cohort} \leftrightarrow \text{matched BI courses}.} \quad (21)$$

7 Historical Data Must Not Be Substituted

A critical methodological principle is that a modern grade distribution cannot be silently substituted for the historical distribution.

If the disputed grade occurred in semester t_0 , then the correct population is

$$\mathcal{G}_{t_0} = \{G_1, \dots, G_n\}_{\text{GRA6539}, t_0}. \quad (22)$$

A contemporary distribution

$$\mathcal{G}_{t_1} \quad (23)$$

is a different statistical population.

Therefore,

$$\boxed{\mathcal{G}_{t_1} \neq \mathcal{G}_{t_0}} \quad (24)$$

unless historical evidence establishes comparability.

This prevents hindsight bias and survivorship bias.

8 The Missing Assessment Record

The most important unresolved empirical object is the contemporaneous assessment record.

Define the record vector

$$R = (S, Q, M, C, E, X, P), \quad (25)$$

where

S = student submission, (26)

Q = examination or assessment questions, (27)

M = marking scheme or rubric, (28)

C = examiner comments, (29)

E = examiner identity and assessment record, (30)

X = component scores, (31)

P = applicable procedural documentation. (32)

The evidentiary hierarchy can then be represented as

$$\boxed{\begin{array}{c} \text{Primary} \\ \text{assessment} > \text{institutional} > \text{secondary} > \text{statistical} \\ \text{record} \quad \text{recollection} \quad \text{reconstruction} \quad \text{inference} \end{array}}. \quad (33)$$

The absence of R cannot prove that the F was improperly awarded.

But it also prevents the historical assessment from being independently reconstructed from primary evidence.

This generates a separate institutional question:

$$\boxed{R = \emptyset \quad \Rightarrow \quad \text{Why is the assessment record unavailable?}} \quad (34)$$

The answer may be entirely benign, such as a lawful retention policy. Alternatively, the record may exist but have become difficult to retrieve. Those possibilities must be distinguished empirically.

9 Ho–Lee as a Mathematical Case Study

The Ho–Lee model is historically important in fixed-income mathematics. Its short-rate dynamics may be represented schematically as

$$dr_t = \theta(t) dt + \sigma dW_t. \quad (35)$$

The model was introduced as an arbitrage-free interest-rate model capable of supporting contingent-claim valuation.

However, its simplicity creates modelling limitations.

In particular, constant volatility does not represent a general stochastic volatility structure:

$$\sigma_t \neq \sigma \quad (36)$$

in a richer term-structure model.

This matters for nonlinear and path-dependent claims because barrier probabilities depend on the dynamics of the underlying state variables.

10 A Necessary Correction Concerning Knock-Out Claims

A strong mathematical statement must distinguish between the Ho–Lee model itself and a complete pricing architecture.

Suppose a claim knocks out when a process S_t enters a barrier region B . Define the survival indicator

$$I_t = \mathbf{1}_{\{\tau_B > t\}}, \quad (37)$$

where

$$\tau_B = \inf\{t \geq 0 : S_t \in B\}. \quad (38)$$

A lattice can then augment the state:

$$X_t = (r_t, I_t). \quad (39)$$

The transition becomes

$$(r_t, I_t) \longrightarrow (r_{t+\Delta t}, I_{t+\Delta t}), \quad (40)$$

where

$$I_{t+\Delta t} = I_t \mathbf{1}_{\{S_{t+\Delta t} \notin B\}}. \quad (41)$$

Consequently, it is too strong to state that a binomial Ho–Lee lattice is mathematically incapable of representing a knock-out state.

The stronger and more defensible proposition is:

Ho–Lee alone does not specify every stochastic state required by an arbitrary path-dependent claim.

(42)

If the barrier depends on another stochastic factor, then that factor and its dependence structure must be specified.

For example,

$$dS_t = \mu_S(t, S_t, r_t) dt + \sigma_S(t, S_t, r_t) dW_t^S. \quad (43)$$

If interest rates and the underlying asset have correlated shocks, then

$$dW_t^S dW_t = \rho dt. \quad (44)$$

A one-factor constant-volatility Ho–Lee specification does not automatically provide this complete multidimensional structure.

11 Lattice and Monte Carlo Demonstration

The computational distinction can be formalized by comparing two models, indexed by $m \in \{HL, R\}$, where HL denotes Ho–Lee and R denotes a richer reference model.

For a vanilla claim,

$$V_m^V = \mathbb{E}_m^Q [D_T \Pi_T]. \quad (45)$$

For a knock-out claim,

$$V_m^{KO} = \mathbb{E}_m^Q [D_T \Pi_T I_T]. \quad (46)$$

It is entirely possible for two models to produce similar vanilla prices,

$$|V_{HL}^V - V_R^V| \approx 0, \quad (47)$$

while producing materially different barrier values,

$$|V_{HL}^{KO} - V_R^{KO}| \gg 0. \quad (48)$$

This is a model-risk phenomenon rather than an algebraic contradiction.

12 Monte Carlo Algorithm

A generic Monte Carlo experiment can be represented by the following pseudocode.

INPUT:

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calibrated yield curve
Ho–Lee parameters
reference-model parameters
barrier B
maturity T
number of paths N
time grid {t_0, ..., t_K}
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FOR each model m:
  FOR j = 1,...,N:
    initialize state X_0
    initialize survival I_0 = 1

    FOR k = 0,...,K-1:
      simulate X_{t_{k+1}} conditional on X_{t_k}

      IF barrier condition is reached:
        set I_{t_{k+1}} = 0
      ELSE:
        set I_{t_{k+1}} = I_{t_k}

    calculate discounted payoff
    store discounted payoff

  calculate sample mean
  calculate Monte Carlo standard error

COMPARE:
  vanilla prices
  knock-out prices
  barrier-hit probabilities
  sensitivity to time-step size
  sensitivity to volatility specification
  sensitivity to correlation assumptions

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The Monte Carlo estimator is

$$\hat{V} = \frac{1}{N} \sum_{j=1}^N D_T^{(j)} \Pi_T^{(j)} I_T^{(j)}. \quad (49)$$

Its standard error is

$$SE(\hat{V}) = \frac{\hat{\sigma}_{\Pi}}{\sqrt{N}}. \quad (50)$$

The computational experiment therefore separates numerical error from model error.

13 Lattice Algorithm

A corresponding lattice procedure is:

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CONSTRUCT time grid
CALIBRATE initial term structure

FOR each time step:
  generate admissible short-rate states

FOR each state:
  determine up/down transitions
  determine discount factors

AUGMENT each state with survival status

FOR terminal nodes:
  calculate payoff if survival = 1
  set payoff = 0 if survival = 0

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BACKWARD INDUCTION:

calculate risk-neutral continuation values

RETURN:

initial-node value

The backward induction equation is

$$V_t = D_{t,t+\Delta t} [p_t V_{t+\Delta t}^u + (1 - p_t) V_{t+\Delta t}^d] . \quad (51)$$

With a knock-out condition,

$$V_t^{KO} = I_t D_{t,t+\Delta t} [p_t V_{t+\Delta t}^u + (1 - p_t) V_{t+\Delta t}^d] . \quad (52)$$

Thus the computational infrastructure needed for a barrier-state extension is conceptually straightforward once the stochastic state is fully specified.

14 Model Risk

The model-risk decomposition can be written as

$$\Delta V = \Delta V_{\text{calibration}} + \Delta V_{\text{dynamics}} + \Delta V_{\text{discretization}} + \Delta V_{\text{path}} + \Delta V_{\text{parameter}} . \quad (53)$$

Here,

$$\Delta V_{\text{calibration}} = \text{difference generated by calibration assumptions}, \quad (54)$$

$$\Delta V_{\text{dynamics}} = \text{difference generated by stochastic dynamics}, \quad (55)$$

$$\Delta V_{\text{discretization}} = \text{difference generated by numerical approximation}, \quad (56)$$

$$\Delta V_{\text{path}} = \text{difference generated by path dependence}, \quad (57)$$

$$\Delta V_{\text{parameter}} = \text{difference generated by parameter uncertainty}. \quad (58)$$

For barrier securities, ΔV_{path} can be particularly important.

15 Why Historical Model Criticism Matters

The historical development of fixed-income modelling demonstrates that model limitations are not unusual.

Ho and Lee introduced their model in the 1980s. Subsequent work, including analysis associated with Dybvig and Steeley, identified limitations and alternative approaches to interest-rate dynamics.

The appropriate intellectual history is therefore not

$$\text{Ho-Lee} \longrightarrow \text{perfect model} \longrightarrow \text{later rejection}. \quad (59)$$

It is more accurately represented as

$$\text{Ho-Lee} \longrightarrow \begin{array}{c} \text{empirical and} \\ \text{theoretical criticism} \end{array} \longrightarrow \begin{array}{c} \text{richer} \\ \text{term-structure} \\ \text{models} \end{array} \longrightarrow \begin{array}{c} \text{continuing} \\ \text{model-risk} \\ \text{research} \end{array}. \quad (60)$$

The existence of earlier criticism does not invalidate a later independent criticism. It does, however, require the later claim to be stated precisely.

16 Binomial Lattices and Computational Feasibility

The binomial option-pricing tradition predates the Ho–Lee model. The canonical Cox–Ross–Rubinstein framework was published in 1979, while discrete-time option-pricing methods were already developing during the preceding decade.

Therefore,

$$\boxed{\begin{array}{c} \text{binomial} \\ \text{computational} < \text{Ho-Lee} \\ \text{methodology} \quad \text{chronology} \end{array}} \quad (61)$$

This is relevant to the historical BI question because the mere absence of modern computing infrastructure would not constitute a satisfactory explanation for an inability to perform a lattice experiment.

The 2013–2014 GRA 6539 documentation itself identifies binomial and Monte Carlo methods as relevant quantitative techniques.

Thus the historically appropriate question is not

$$\text{Could BI possibly have performed such a calculation?} \quad (62)$$

but

$$\boxed{\text{Was such a calculation actually performed in the assessment?}} \quad (63)$$

17 Institutional Quality Assurance

A university assessment system is a socio-technical system. Its reliability depends on people, rules, data, computational tools, records, governance, and review.

A stylized institutional reliability function is

$$R_{\text{institution}} = f(R_{\text{people}}, R_{\text{process}}, R_{\text{data}}, R_{\text{computation}}, R_{\text{records}}, R_{\text{governance}}). \quad (64)$$

A failure in one component does not automatically imply institutional collapse. But an inability to reconstruct an important historical decision creates a traceability problem.

The institutional audit therefore asks:

$$Q_1 : \text{Was the assessment criteria documented?} \quad (65)$$

$$Q_2 : \text{Was the assessment performed according to those criteria?} \quad (66)$$

$$Q_3 : \text{Was the result independently reviewed where required?} \quad (67)$$

$$Q_4 : \text{Was the assessment record retained appropriately?} \quad (68)$$

$$Q_5 : \text{Can the institution reconstruct the decision?} \quad (69)$$

18 Accreditation Perspective

AACSB, EQUIS, and AMBA/BGA are not substitutes for an institution’s examination records or Norwegian legal mechanisms. Their relevance lies in institutional quality assurance.

A useful abstraction is:

$$\boxed{\text{Accreditation} = \begin{array}{c} \text{external} \\ \text{quality} \\ \text{assurance} \end{array} + \begin{array}{c} \text{institutional} \\ \text{accountability} \end{array} + \begin{array}{c} \text{continuous} \\ \text{improvement} \end{array}} \quad (70)$$

The existence of accreditation does not prove that every historical assessment was correct.

Conversely, one disputed grade does not establish accreditation failure.

The relevant institutional question is therefore:

Can the school demonstrate that its assessment system is documented, traceable, reliable, and capable of detecting and correcting errors?	(71)
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This is an assurance-of-learning and governance question rather than a request for an accreditor to overturn an individual grade.

19 Data Science Extension

The complete audit can be represented as a supervised anomaly-detection problem.

Let

$$Y_i = \text{observed grade.} \quad (72)$$

Let the feature vector be

$$X_i = (\text{prior grades, course, semester, assessment type, examiner, programme, cohort}). \quad (73)$$

A conditional anomaly score can then be defined as

$$A_i = -\log P(Y_i | X_i). \quad (74)$$

Large values of A_i identify observations that are unusual conditional on their explanatory variables.

A machine-learning implementation could compare logistic regression, ordinal regression, random forests, gradient boosting, or Bayesian hierarchical models.

However, the statistical model must not be used to manufacture a causal conclusion.

The proper sequence is:

anomaly detection $\rightarrow$ human review $\rightarrow$ primary-document verification.	(75)
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20 Matched-Control Design

A particularly useful institutional design is to construct two corpora:

$$\mathcal{B} = \text{assessments associated with the person under investigation,} \quad (76)$$

$$\mathcal{C} = \text{matched assessments without that association.} \quad (77)$$

The groups should be matched by course type, discipline, assessment type, semester, programme, and student population wherever possible.

For a measurable outcome Y , define

$$\Delta = E[Y | \mathcal{B}] - E[Y | \mathcal{C}]. \quad (78)$$

Possible outcomes include:

- grade distributions;
- failure rates;
- dispersion of grades;
- frequency of explicit uncertainty analysis;
- frequency of model sensitivity analysis;
- number of assessment components;
- proportion of assessments involving quantitative modelling;
- record completeness;

- frequency of formal complaints;
- time required to reconstruct historical assessments.

A statistically significant Δ would justify further examination, but would still require substantive interpretation.

21 Food and Environmental Statistical Audit

The same methodology can be applied to public scientific assessments involving food and environmental risk.

This is particularly relevant to the current professional environment of Barbara Bukhvalova, whose publicly identified work at the Norwegian Scientific Committee for Food and Environment includes statistics and modelling.

The appropriate scientific corpus consists of publicly available VKM reports rather than private personnel information.

Relevant classes include:

- food contamination;
- exposure assessment;
- food-consumption modelling;
- benefit-risk analysis;
- environmental exposure;
- pesticide risk;
- ecological and epidemiological modelling;
- uncertainty analysis;
- Monte Carlo simulation;
- statistical treatment of observations below analytical limits.

The audit should test methodology rather than assume misconduct.

For an exposure variable X , for example, the assessment may depend on

$$E[X] = \int x f(x) dx. \quad (79)$$

If the distribution is highly skewed, the risk assessment may depend strongly on tail behaviour:

$$P(X > x^*) = 1 - F(x^*). \quad (80)$$

Consequently, alternative distributional assumptions can produce different risk estimates:

$$\Delta R = R(f_1) - R(f_2). \quad (81)$$

This provides a principled way to investigate whether conclusions are robust.

22 Grilled Food and Food-Consumption Assessments

VKM's assessment of grilled and barbecued food provides a natural statistical case study because food-contaminant assessments can involve censored observations, limits of quantification, consumption frequencies, scenario assumptions, distributions, and simulated exposures.

A generic Monte Carlo exposure model is

$$E_j = C_j Q_j F_j, \quad (82)$$

where C_j represents concentration, Q_j quantity consumed, and F_j frequency or scaling factors.

The aggregate exposure can be represented as

$$E = \sum_{j=1}^J E_j. \quad (83)$$

The resulting risk estimate may depend on the treatment of observations below the limit of quantification.

If two legitimate censoring assumptions are represented by f_1 and f_2 , then

$$\Delta E = E_{f_1}[E] - E_{f_2}[E]. \quad (84)$$

A robustness audit should therefore report both the baseline result and reasonable alternative specifications.

23 Fish in the Norwegian Diet

Benefit-risk analysis introduces an additional complication: positive and negative effects must be represented simultaneously.

Let

$$B = \text{expected health benefit}, \quad (85)$$

$$R = \text{expected health risk}. \quad (86)$$

A simplified net-benefit criterion is

$$NB = B - R. \quad (87)$$

The sign of NB can be sensitive to assumptions concerning consumption, contaminant exposure, biological effects, and population heterogeneity.

Therefore the appropriate audit asks not merely for the point estimate but for

$$\frac{\partial NB}{\partial \theta_k}, \quad (88)$$

where θ_k denotes a modelling assumption.

A large derivative indicates high sensitivity to that assumption.

24 Environmental Modelling and Norwegian Conditions

Environmental risk modelling provides a related methodological principle: models developed for one physical environment may not automatically be representative in another.

A generic environmental model can be represented as

$$Y = g(X, \theta, E), \quad (89)$$

where X represents observations, θ model parameters, and E environmental conditions.

If a model calibrated under environmental regime E_1 is applied to E_2 , then

$$g(X, \theta, E_1) \neq g(X, \theta, E_2) \quad (90)$$

may hold materially.

This is the environmental analogue of the fixed-income model-risk problem: **model validity is conditional on the state space in which the model is used.**

25 Scientific and Philosophical Dimensions

The philosophy of science requires distinguishing falsification, anomaly, model inadequacy, and theory replacement.

A scientific anomaly is an observation

$$O \notin \mathcal{R}(M), \quad (91)$$

where $\mathcal{R}(M)$ denotes the range of outcomes regarded as plausible under model M .

However,

$$O \notin \mathcal{R}(M) \quad (92)$$

does not uniquely identify the cause.

Possible explanations include:

- measurement error;
- sampling variation;
- model misspecification;
- parameter uncertainty;
- omitted variables;
- procedural error;
- genuine structural change.

This is why the correct response to an anomaly is investigation rather than automatic attribution.

26 Psychology and Human Decision-Making

Assessment is also a human decision process.

Potential cognitive mechanisms include:

- anchoring;
- confirmation bias;
- status-quo bias;
- authority bias;
- attribution error;
- motivated reasoning;
- hindsight bias.

A rigorous assessment system therefore attempts to reduce the influence of individual cognitive biases through explicit criteria, independent review, standardized procedures, and documentation.

This yields a governance principle:

Transparent procedure > reliance on individual judgement alone.	(93)
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The principle applies equally to academic grading, financial risk management, scientific risk assessment, and public administration.

27 Economics of Institutional Reputation

Institutional reputation can be represented as a capital stock.

Let K_t denote institutional reputational capital. Then

$$K_{t+1} = K_t + I_t - D_t, \quad (94)$$

where I_t represents reputation-enhancing information and D_t represents reputation-damaging information.

Transparent correction of a genuine error may reduce short-term reputational cost while increasing long-term institutional credibility.

Thus,

$$\text{short-run disclosure cost} < \text{expected long-run cost of concealed error} \quad (95)$$

may hold when the probability and consequences of later discovery are large.

This logic applies to universities, financial institutions, pension funds, regulators, scientific committees, and governments.

28 Pension and Financial-System Relevance

Interest-rate models have applications in:

- pension liability valuation;
- insurance;
- bank asset-liability management;
- derivatives pricing;
- hedging;
- fixed-income portfolio management;
- regulatory stress testing;
- collateral and funding analysis.

The general model-risk principle is

$$\boxed{\text{model error} \times \text{economic exposure} \times \text{nonlinearity} = \text{potential systemic significance.}} \quad (96)$$

However, a mathematical limitation of Ho–Lee does not establish that Ho–Lee caused pension crises worldwide.

That causal proposition would require an entirely different empirical study involving pension-fund data, asset allocations, liability structures, hedging strategies, model usage, and counterfactual analysis.

29 Welfare Economics

The social cost of an institutional error depends not only on the private cost to the individual but also on externalities.

Let

$$W = U_{\text{student}} + U_{\text{institution}} + U_{\text{society}}. \quad (97)$$

A defective assessment process can impose private costs on students and potentially wider costs through educational selection, professional placement, institutional trust, and human-capital allocation.

The welfare objective is therefore not simply to maximize the probability of correct grading in one case, but to minimize expected system-wide assessment error:

$$\min E[L_{\text{assessment}}]. \quad (98)$$

30 Political Science and Institutional Governance

Institutions can be modelled as systems of delegated authority.

The chain is approximately

$$\text{Student} \rightarrow \text{Examiner} \rightarrow \text{Department} \rightarrow \text{Institution} \rightarrow \text{Accreditor} \rightarrow \text{Public Governance.} \quad (99)$$

Each additional layer should increase accountability rather than merely increase distance from the original decision.

A robust governance architecture therefore requires

$$\text{Authority} + \text{Documentation} + \text{Review} + \text{Appeal} + \text{Auditability.} \quad (100)$$

31 International Relations and State Capacity

The same architecture appears in state institutions.

A state with high formal authority but weak information preservation can exhibit

$$\text{Authority} \gg \text{Traceability.} \quad (101)$$

Conversely, a state with strong recordkeeping and independent review can exhibit

$$\text{Authority} \approx \text{Traceability.} \quad (102)$$

For this reason, historical records are not merely administrative artefacts. They constitute part of institutional state capacity.

32 Warfare Economics and Strategic Resilience

The same principle appears in military systems.

A command structure that cannot reconstruct the origin of a decision cannot reliably distinguish:

- successful doctrine;
- accidental success;
- procedural failure;
- intelligence failure;
- equipment failure;
- operator error.

A resilient organization therefore maintains an after-action record.

The general principle is

$$\boxed{\text{traceability} \rightarrow \text{learning} \rightarrow \text{adaptation} \rightarrow \text{resilience.}} \quad (103)$$

This is equally applicable to universities and scientific institutions.

33 Computer Science Interpretation

The assessment process can be interpreted as a distributed information system.

Each assessment generates an event:

$$e_i = (\text{student}, \text{course}, \text{assessment}, \text{examiner}, \text{timestamp}, \text{result}, \text{evidence}). \quad (104)$$

An auditable system should satisfy the property

$$\forall e_i, \quad \exists \text{recoverable evidence}(e_i). \quad (105)$$

If that property fails, reproducibility fails.

This resembles a database integrity constraint:

$$\boxed{\text{assessment result} \leftrightarrow \text{supporting record}.} \quad (106)$$

The absence of that linkage creates an auditability gap.

34 Artificial Intelligence and Automated Audit

The framework can be generalized into an AI-assisted audit system.

A model can classify historical assessments according to

$$\hat{y}_i = f_{\theta}(X_i), \quad (107)$$

where $\hat{y}_i$ represents the expected grade distribution conditional on observable variables.

An anomaly score can then be defined as

$$A_i = |Y_i - \hat{y}_i|. \quad (108)$$

A Bayesian version can instead calculate

$$P(Y_i | X_i, \mathcal{D}), \quad (109)$$

where $\mathcal{D}$ denotes the historical dataset.

AI should not be the final adjudicator.

The proper architecture is

$$\boxed{\text{AI detection} \rightarrow \text{statistical validation} \rightarrow \text{human review} \rightarrow \text{primary-record verification}.} \quad (110)$$

35 Unified Audit Architecture

The entire investigation can be summarized in the following architecture.

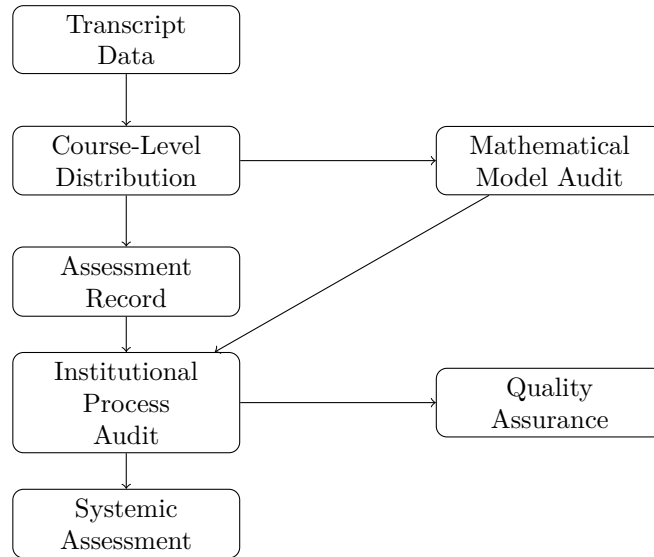


Figure 1: Evidence-driven assessment and institutional audit architecture.

The architecture deliberately separates the statistical, mathematical, and institutional branches before recombining them.

36 Evidence Matrix

Table 2: Current evidentiary status

Question	Status	Interpretation
F is unusual within the transcript	Established	Z_F is approximately -2.3 to -2.4 .
GRA 6539 had quantitative modelling content	Established	Historical BI documentation explicitly identifies relevant modelling and computational methods.
BI could conceptually implement a lattice/Monte Carlo test	Established	The required methodology existed and was explicitly part of the course framework.
H2014 course-level F anomaly	Unresolved	Requires the actual historical grade distribution.
Specific examiner calculation producing the F	Unresolved	Requires the contemporaneous assessment record.
Improper grading	Unestablished	Cannot be inferred solely from the transcript anomaly.
Institution-wide assessment failure	Unestablished	Requires evidence of recurrence or systemic process failure.
Worldwide pension-fund causation by Ho-Lee	Unestablished	Requires independent financial and causal evidence.

37 Formal Audit Protocol

The recommended protocol is:

INPUT:

```
historical transcript
historical course documentation
historical grade distributions
assessment records
examiner documentation
comparable courses
```

STEP 1:

```
calculate transcript-level anomaly statistics
```

STEP 2:

```
retrieve the exact historical course cohort
```

STEP 3:

```
calculate course-level grade statistics
```

STEP 4:

```
compare with matched courses
```

STEP 5:

```
reconstruct the assessment process
```

STEP 6:

```
identify mathematical propositions actually assessed
```

STEP 7:

```
reproduce relevant calculations
```

STEP 8:

compare alternative models

STEP 9:

conduct sensitivity and robustness analysis

STEP 10:

distinguish individual error from systemic pattern

OUTPUT:

statistical finding
mathematical finding
procedural finding
institutional finding
uncertainty statement

The protocol prohibits causal conclusions before the evidence required for those conclusions exists.

38 A Formal Proposition on Record-Based Accountability

Proposition 1. Suppose a historical assessment result G is challenged and the institution cannot produce the contemporaneous assessment record R . Then the absence of R prevents direct verification of the original assessment procedure, although it does not logically imply that the assessment was improper.

Proof. Direct verification requires evidence of the relevant assessment inputs, criteria, examiner reasoning, and resulting grade. If R is unavailable, at least one of those evidentiary components cannot be directly inspected. Therefore direct historical verification is incomplete. However, the proposition

$$R = \emptyset \tag{111}$$

does not logically imply

$$\text{improper assessment.} \tag{112}$$

Hence the absence of R establishes an evidentiary limitation but not the causal conclusion. $\square$

39 A Formal Proposition on Transcript Anomaly

Proposition 2. Under the numerical grade coding adopted above, the observed F is a substantial negative outlier within the student's transcript distribution.

Proof. The numerical mean is approximately 3.308, while the F corresponds to zero. The population standard deviation is approximately 1.380. Consequently,

$$Z_F = \frac{0 - 3.308}{1.380} \approx -2.40. \tag{113}$$

Therefore the F is approximately 2.4 standard deviations below the transcript mean. The sample-standard-deviation formulation yields approximately -2.30 . Thus the conclusion that the F is a substantial transcript-level statistical outlier follows from the specified coding and observed grades. $\square$

40 A Formal Proposition on Ho–Lee and Path Dependence

Proposition 3. A Ho–Lee short-rate process can be incorporated into a computational architecture containing a knock-out state, but the Ho–Lee specification alone does not uniquely determine every stochastic process required by an arbitrary path-dependent claim.

Proof. The survival indicator I_t can be introduced as an additional state variable, so a lattice can track whether a barrier has been crossed. Therefore a computational barrier state is possible.

However, if the barrier depends on a process S_t whose dynamics are not fully specified by the Ho–Lee short-rate process, then the law of S_t must be specified separately. Consequently, the Ho–Lee process alone does not uniquely determine the complete path-dependent valuation problem. $\square$

41 Conclusion

The investigation produces a hierarchy of findings rather than a single verdict.

First, the historical transcript contains a statistically unusual observation:

$$\boxed{Z_F \approx -2.3 \text{ to } -2.4.} \quad (114)$$

Second, the historical GRA 6539 documentation establishes that quantitative interest-rate modelling, fixed-income options, binomial trees, Monte Carlo simulation, and analysis extending beyond explicitly covered material were within the stated educational framework.

Third, the mathematical proposition concerning Ho–Lee must be stated carefully. Ho–Lee is not mathematically incapable of being embedded within a knock-out lattice. The stronger issue concerns the completeness of the stochastic state specification, volatility assumptions, path dependence, and model risk.

Fourth, the decisive unresolved issue is not whether an anomaly exists at the transcript level. It does. The decisive unresolved issue is whether the historical course-level and assessment-level evidence corroborates or contradicts the anomaly.

The resulting hierarchy is:

$$\boxed{\text{Transcript anomaly} \rightarrow \text{Course anomaly} \rightarrow \text{Assessment anomaly} \rightarrow \text{Institutional anomaly}} \quad (115)$$

with each arrow requiring additional evidence.

The broader lesson is interdisciplinary. The same principles govern academic assessment, financial modelling, food-safety assessment, environmental science, pension risk, banking, regulatory analysis, and state administration:

$$\boxed{\text{Models must be tested, decisions must be traceable, and anomalies must be investigated without prejudging their cause.}} \quad (116)$$

The ultimate institutional standard is therefore not perfection. It is reproducibility, traceability, correction, and continuous improvement.

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Glossary

Anomaly

An observation that is unusually distant from an appropriate reference distribution or expected conditional outcome.

Assessment Record

The contemporaneous collection of examination materials, submitted work, marking criteria, examiner information, component scores, comments, and procedural documentation needed to reconstruct an assessment.

Barrier

A boundary such that crossing it changes the contractual payoff, including termination of a knock-out claim.

Binomial Lattice

A discrete-time tree representation of possible states used for numerical valuation of financial securities.

Ho–Lee Model

A one-factor short-rate model introduced by Thomas Ho and Sang-Bin Lee for arbitrage-free interest-rate modelling.

Monte Carlo Simulation

A numerical method that estimates expectations by generating repeated random samples from specified stochastic processes.

Model Risk

The possibility that decisions based on a mathematical or statistical model are materially affected by model misspecification, incorrect assumptions, parameter uncertainty, or implementation error.

Path Dependence

Dependence of a security's payoff on the historical trajectory of one or more state variables rather than solely on their terminal values.

Quality Assurance

The institutional processes used to establish, monitor, evaluate, and improve the quality and integrity of academic programmes and their assessment.

Statistical Outlier

An observation unusually distant from a reference distribution according to a specified statistical criterion.

Traceability

The ability to connect an institutional decision to the evidence, procedures, people, calculations, and records that generated it.

Uncertainty Analysis

The systematic examination of how uncertainty in data, parameters, model structure, or assumptions affects an estimated outcome.

Z-score

A standardized measure of an observation's distance from a distribution mean, expressed in standard-deviation units.

The End